

Mini-HPC and Hybrid HPC-Big Data Clusters Project

Introduction

This project aims to demonstrate the setup and execution of both a High-Performance Computing (HPC) cluster and a Hybrid HPC-Big Data cluster. The primary goals include understanding the fundamentals of distributed computing, setting up virtualized environments, implementing distributed machine learning tasks using MPI (Message Passing Interface) for traditional HPC, and leveraging Docker Swarm with Apache Spark for Big Data processing. The project also involves running and evaluating distributed models on a real-world bioinformatics gene expression dataset, showcasing practical applications of these technologies.

Methodology

Virtual Machine Setup and Networking

The foundation of this project involved setting up a virtual 3-node cluster using VirtualBox. This cluster consisted of one master node and two worker nodes, each configured with Ubuntu 24.04.1 LTS. A host-only network was established within VirtualBox to facilitate private communication between the virtual machines, and static IP addresses were assigned to each VM. Passwordless SSH was configured from the master to the worker nodes to enable seamless command execution and file transfer across the cluster, which is crucial for distributed computing environments.

MPI Implementation (Task 1)

For the traditional HPC component, the Message Passing Interface (MPI) was utilized. OpenMPI was installed on all nodes of the cluster, along with Python 3 and the `mpi4py` library, which provides MPI bindings for Python. Two distributed machine learning tasks were implemented:

1. **Distributed Digits Dataset Classification:** A

`distributed_mnist.py` script was developed to classify the scikit-learn digits dataset. The dataset was distributed among the MPI processes, and each process trained a `RandomForestClassifier` on its local data chunk. The local scores were then gathered at the root process (rank 0) to calculate an overall average accuracy.

2. **Distributed Bioinformatics Gene Analysis:** A

`distributed_gene_analysis.py` script was created to perform a similar distributed classification task on a simulated `leukemia_expression.csv` dataset. This demonstrated the application of HPC techniques to real-world bioinformatics problems, where large datasets are common. The methodology mirrored the digits classification, with data partitioning and distributed model training.

Docker Swarm and Spark Implementation (Task 2)

The hybrid HPC-Big Data component involved setting up a Docker Swarm cluster and deploying Apache Spark. Docker was installed on all virtual machines, and a Swarm was initialized on the master node, with the worker nodes joining the Swarm. A `spark-stack.yml` Docker Compose file was used to define and deploy the Spark master and worker services across the Swarm. This setup provided a scalable and fault-tolerant environment for Big Data processing.

For the distributed Spark job, a `distributed_gene_expression_analysis.py` PySpark script was developed. This script read the `leukemia_expression.csv` dataset, utilized Spark MLlib for feature vector assembly, and trained a `RandomForestClassifier`. The model's accuracy was evaluated on a test dataset, showcasing Spark's capabilities in handling large-scale machine learning tasks within a distributed environment.

Conclusion

This project successfully established both a Mini-HPC cluster and a Hybrid HPC-Big Data cluster, demonstrating the fundamental principles and practical applications of distributed computing. Through the implementation of MPI-based machine learning tasks, we explored traditional HPC paradigms for parallel processing. The integration of Docker Swarm with Apache Spark showcased a modern approach to handling Big Data analytics and machine learning in a distributed fashion. The project reinforced the understanding of distributed computing concepts, including data partitioning, inter-process communication, and the deployment of scalable services.